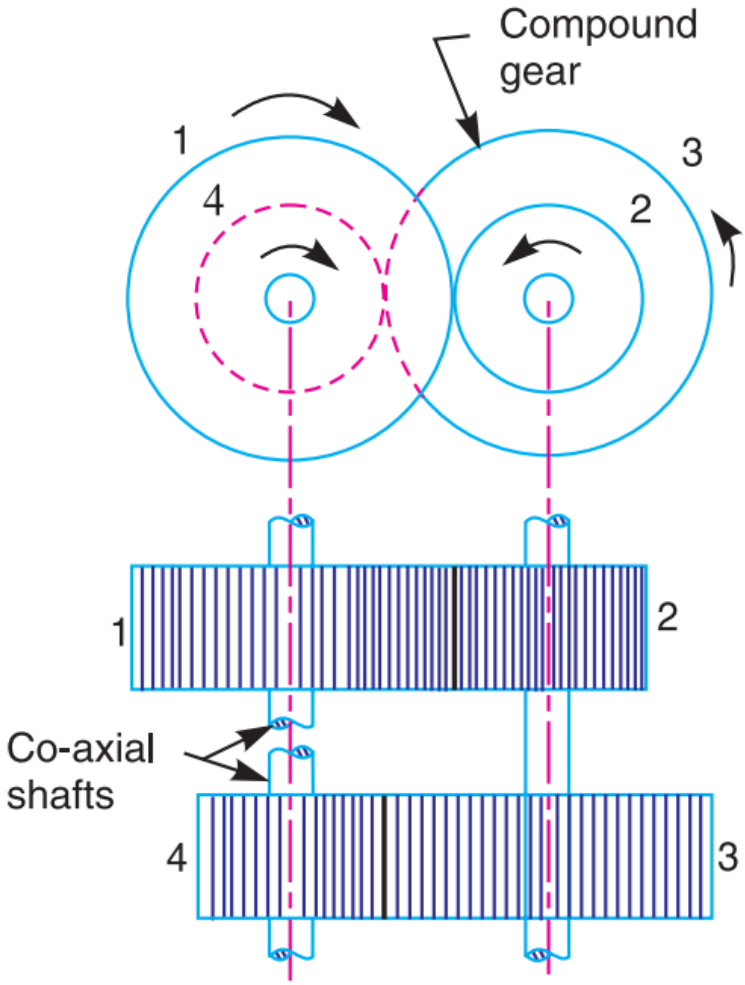
MANIPAL INSTITUTE OF TECHNOLOGY
MANIPAL
(A constituent unit of MAHE, Manipal)

DEPARTMENT OF AERONAUTICAL AND AUTOMOBILE ENGINEERING			
END SEMESTER EXAMINATION		**DECEMBER, 2024**	
Level	**B.Tech**	**Curriculum**	**2022**
Program Name	**Open Elective**	**Course Type**	**C/FC/PE/OE**
Course Name	**Introduction to Farm Machines & Earthmoving Equipment**	**Course Code**	**AAE4305**
Semester	**VII**	**Max Marks**	**50**
Date	**02/12/2024**	**Duration**	**3 Hours**

Instructions:

- **Answer ALL the questions.**
- **Data not provided may be suitably assumed**

Q. No.	Question	Marks	NBA		IET*		BL
			CO	PO	CLO	AHEP4 LO	
1A.	Describe the requirements of the lubrication system.	2	1	1	1	1,4,5,18	3
1B.	Describe the construction of scrapper.	4	3	1	1	1,4,5	3
1C.	List the advantages of disc plough over others.	4	5	7	7	4,5,7	3
2A	Explore the usage of connecting rod in any engine.	2	2	1	1	4,5	2
2B	Explore the operational differences of Articulated type, and rigid frame type loaders.	4	4	3	3	4,5	3
2C	Describe the process of tillage with both the types.	4	5	1	1	4,5	2
3A	Illustrate the standards on which a dozer is specified.	3	3	3	3	2,4,5	3
3B	Differentiate between crawler mounted and truck mounted excavators.	3	4	3,5	3,5	1,4,5	2
3C	Describe the influence of moisture on compaction, and principle of compaction.	4	3	1	1	1,4,5	2
4A	Suggest the work cycle improvement of a bulldozer in between operation, and maintenance.	3	3	5,6	5,6	2,4,5,7	4
4B	Explore the difference between bottom dump and front dump buckets of a shovel.	3	4	3	3	1,4,5	3

4C	Explore the types of Harrows.	**4**	**5**	**1**	**1**	**1,4,5**	**2**
5A	What are the types of work a grader do?	**3**	**3**	**6**	**6**	**1,4,5**	**2**
5B	figure below the primary gear 1 moves with 1200rpm in clockwise, and the number of teeth on gears 1, 2, 3, and 4 are 64, 96, 32, and 48 respectively. Find the speed and direction of the gear 4. 	**3**	**4**	**2**	**2**	**2,4,5**	**3**
5C	Explore the factors on which classification and specifications of rippers depend.	**4**	**4**	**3**	**3**	**1,4,5**	**2**

Retain only one. Delete the others

xxxxxxx Fill in appropriately.

**** Applicable for only IET-accredited courses (modules). The IET columns may be deleted for non-IET courses.***

MARK DISTRIBUTION (CO-wise & BTL-wise)

CO / CLO	Total Marks
1	***2***
2	***2***
3	***17***
4	***17***
5	***12***

BL	Total Marks
2	24
3	20
4	6
5	